ArA-DF 2026: Findings of the First Shared Task on Arabic Speech Deepfake Detection

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Abstract

We present ArA-DF 2026, the first shared task for Arabic speech deepfake detection, held at ArabicNLP 2026. Unlike existing benchmarks that conflate all distribution shift into a single evaluation set, ArA-DF 2026 separates dialect generalization (Track 1: unseen Arabic dialects and speakers) from acoustic robustness (Track 2: codec compression, noise, reverberation, re-encoding). We release 43,500 labeled training utterances spanning MSA and regional dialects, plus blind evaluation sets of 144,210 (Track 1) and 127,746 (Track 2) utterances, ranking systems by EER. Fourteen teams competed on Track 1 and twelve on Track 2, with eight system description papers submitted. The best submissions cut EER from 14.54% to 0.87% (Track 1) and from 27.11% to 0.66% (Track 2). Across independent submissions, three findings recur: augmentation and data curation move performance far more than architecture; simpler pooling heads often outperform graph-attention back-ends once augmentation is applied; and external synthetic Arabic speech with mismatched generation artifacts hurts rather than helps. Residual errors concentrate on short utterances and on spoofed audio misclassified as bona fide.

1 Introduction

Neural text-to-speech (TTS) and voice conversion (VC) have reached the point where synthetic speech is difficult to distinguish from genuine recordings by ear. In a controlled listening study, human listeners identified deepfake audio correctly only 73% of the time, and familiarizing them with examples improved performance only marginally (Mai et al., 2023); a large-scale evaluation reached similar conclusions (Warren et al., 2024). The same technologies that enable accessible speech interfaces (Tan, 2023; Chen et al., 2025) also enable impersonation, voice fraud and the fabrication of audio evidence (Li et al., 2025), which makes reliable automatic detection a practical safeguard rather than an academic exercise.

The research community has responded with a decade of organized evaluation, from the ASVspoof series (Wu et al., 2017) through the single-language and single-domain challenges that followed (Figure 1). That work has produced a stable recipe of a self-supervised (SSL) front end, a lightweight back end and heavy augmentation (Section 2). Despite this convergence, the field's central problem remains open: detectors that are near-perfect in-domain degrade sharply on unseen synthesis methods, unseen channels and unseen recording conditions (Müller et al., 2022; Liu et al., 2023; Wang et al., 2024). Recent large-scale comparisons make the point quantitatively, across 400 trained systems, the choice of pre-trained front end and of training corpus dominates the variance in cross-domain performance, while the back-end classifier contributes less than one point of EER (El Kheir et al., 2026a).

Almost all of this evidence comes from English. Arabic is spoken by roughly 450 million people, is served by a growing set of commercial and open Arabic TTS and VC systems, and presents a form of variation that English benchmarks do not exercise: Modern Standard Arabic (MSA) and the regional dialects differ substantially in phonology, morphology and prosody, so that a detector trained on one variety may not transfer to another. Arabic-specific resources have begun to appear (Section 2), but before this task there was no shared, blind benchmark on which Arabic detectors could be compared under a common protocol.

ArA-DF 2026 fills that gap. Its distinguishing design choice is that it varies dialect and acoustics *independently* across two tracks rather than pooling them. This matters because the two failure modes have the same symptom, an accepted spoof, but different remedies, and a single aggregate score cannot tell an organizer or a practitioner which

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 1473–1485
October 28-29, 2026 ©2026 Association for Computational Linguistics

one they are looking at. The separation proved informative: the two tracks rank systems differently and reward different designs (Section 7). Our contributions are: (1) **task design**, two tracks that isolate dialect shift from acoustic shift under a single labeling convention and metric; (2) **dataset release**, a labeled Arabic bona fide/spoofed corpus with dialect-family and gender metadata, plus blind evaluation partitions of 144,210 and 127,746 utterances; (3) **baseline and infrastructure**, a public XLS-R+AASIST baseline, scoring code, and two Codabench leaderboards; and (4) **findings**, an analysis of eight system description papers identifying what actually moved the needle, plus reproducible *negative* results on external synthetic data that we believe are as useful as the positive ones.

2 Related Work

Benchmarks and challenges. Figure 1 places ArA-DF 2026 in the lineage of organized evaluation. The ASVspoof series (Kinnunen et al., 2017; Wang et al., 2020; Liu et al., 2023; Wang et al., 2024) established the conventions the present task inherits: a binary bona fide/spoofed decision, threshold-free scoring by equal error rate, and a blind evaluation partition; ASVspoof 2021's codec and telephony condition definitions were reused directly by several ArA-DF 2026 participants. The ADD challenges (Yi et al., 2022, 2023) narrowed organized evaluation to a single language, Mandarin, the branch this task extends to Arabic. For generalization specifically, the In-the-Wild corpus (Müller et al., 2022) showed that detectors trained on controlled data collapse on uncurated audio, MLAAD (Müller et al., 2024) extended the problem past 50 languages, and SpoofCeleb (Jung et al., 2025) scaled it to thousands of speakers. Aggregation efforts such as Speech-DF-Arena (Dowerah et al., 2026) and AUDDT (Zhu et al., 2025) place many of these sets behind a single interface, and report that AASIST-based systems generalize comparatively poorly out of domain (see Section 8).

Front ends, back ends and augmentation. Hand-crafted spectral features (Todisco et al., 2017; Yang et al., 2019) have largely given way to SSL encoders: wav2vec 2.0 (Baevski et al., 2020), its cross-lingual extension XLS-R (Babu et al., 2022; El Kheir et al., 2025a), WavLM (Chen et al., 2022), HuBERT (Hsu et al., 2021) and W2V-BERT-2.0 (Barrault et al., 2023). Layer-wise analyses show that intermediate layers carry more spoofing-relevant information than the final one (El Kheir et al., 2025c; Xiao and Vu, 2025), motivating the learnable weighted-sum aggregation that most ArA-DF 2026 systems adopt; fine-tuning such an encoder end-to-end with heavy augmentation is now the reference recipe (Tak et al., 2022b). On the back-end side, AASIST (Jung et al., 2022) remains the default, alongside RawNet2 (Tak et al., 2021), ECAPA-TDNN (Desplanques et al., 2020), attentive statistics pooling (Okabe et al., 2018), temporal-channel modeling (Truong et al., 2024), Nes2Net (Liu et al., 2025) and BiCrossMamba-ST (Kheir et al., 2025). Augmentation is the third pillar: RawBoost (Tak et al., 2022a), room impulse responses (Ko et al., 2017) and MUSAN noise (Snyder et al., 2015) are near-universal, and post-training on large artifact-bearing corpora has recently been shown to improve transfer (Ge et al., 2025). DeepFense (El Kheir et al., 2026a) packages these components as interchangeable modules; two ArA-DF 2026 teams built their pipelines on it.

Arabic speech and Arabic anti-spoofing. Arabic speech technology has grown rapidly, from dialect-aware recognition and large multi-dialect corpora such as SADA (Alowisheq et al., 2024) to open Arabic speech synthesis (Toyin et al., 2025). Arabic anti-spoofing is newer: ArFake (Elsetohy et al., 2025) introduced a multi-dialect benchmark with leave-one-generator-out and leave-one-dialect-out protocols targeting the same two generalization axes as this task; AFAD-MSA (Issa, 2026) released MSA-focused fake-audio data; and MLAAD (Müller et al., 2024) includes an Arabic split. These resources informed systems from several of our participants. ArA-DF 2026 builds on this momentum, following prior ArabicNLP shared tasks such as Iqra'Eval (El Kheir et al., 2025b), by contributing a hosted benchmark with a large blind evaluation partition, an open baseline and tutorial, a live leaderboard, and two tracks that vary dialect and acoustic condition independently.

3 Task Motivation and Overview

3.1 Motivation and Task Definition

Zero-shot TTS and voice cloning APIs are available in abundance making impersonation, fraud, and fabricating audio evidence much easier than ever. Among anti-spoofing benchmarks, Arabic remains under-represented despite a demanding mix of MSA and regional dialects that differ in

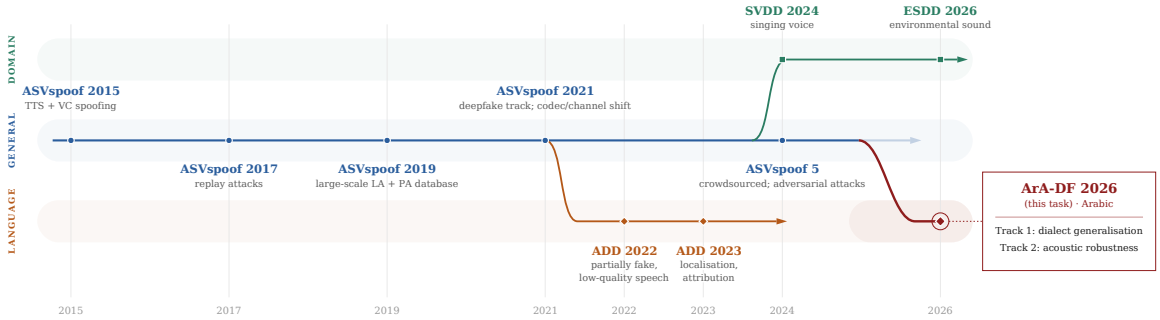


Figure 1: Timeline of speech deepfake detection challenges, grouped by benchmark scope: general-purpose (the English-dominant ASVspoof series), single-language (ADD, Mandarin; ArA-DF 2026, Arabic, in red), and single-domain (SVDD (Zhang et al., 2024), ESDD (Yin et al., 2025)).

phonology, prosody and lexical patterns. We therefore position ArA-DF 2026 as, to the best of our knowledge, the first dedicated blind shared task for Arabic speech deepfake detection.

We have chosen binary utterance classification (bonafide or spoofed) over generator attribution to lower the entry barrier while retaining questions of calibration and robustness both acoustically and towards unseen dialects. Systems receive 16-kHz mono speech; label `1` is bona fide and `0` is synthetic. System submissions include `audio_id,logit`, where larger logits favor bona fide speech. In order to rank systems, we use Equal Error Rate (EER) (Section 5) as the metric, so that there is no common threshold required. The spoofed split of the final benchmark contains TTS spoofs, and not identified voice-conversion or partial-spoof category.

3.2 Tracks

Track 1: Dialect Generalization and Beyond The first track of the challenge is to measure the robustness to dialects and synthesis conditions beyond what is represented in supervised training data. The training and development splits provide MSA, Egyptian, and Gulf Arabic as the dialect labels along with the bona fide or spoof binary label. In the testing phase, Track 1 additionally introduces Levantine and Maghrebi samples and held-out generation methods. Every test sample also receives one telephony-style logical-access (LA) transform and one distribution-format (DF) transform. Track 1, therefore is deliberately a joint dialect, generator, and channel-shift problem, rather than measuring the robustness of dialect alone (though it is our main focus).

Track 2: Acoustic Robustness. The second track retains the three core dialect families (MSA, Egyptian and Gulf) and probes whether the systems are robust to downstream acoustic changes. Approximately 20% of its tests are from clean signal, while the remainder is divided nearly equally among lossy coding, additive environmental noise, and room reverberation. Held-out synthesis systems also occur, so errors may reflect the acoustic shift, the generator shift, or their interaction. This reflects the usual deployment setting, where synthetic audio is commonly compressed, transmitted, or re-encoded prior to analysis. The train/development splits are common across tracks. The evaluation uses separate CodaBench competitions. Teams could enter either or both. Because their processed tests derive substantially from common sources, cross-track scores are complementary rather than independent experiments or controlled estimates of channel effects.

4 Dataset and Resources

4.1 Release Snapshot

The public Hugging Face repository (Arabic-Speech, 2026) is the authoritative release. Its six canonical partitions contain 345,672 utterances (884.88 hours). Four post-challenge labeled configurations mirror the test metadata and add gold labels without duplicating audio. All recordings are 16-kHz mono PCM-16 FLAC, packaged in WebDataset TAR shards (Breuel, 2020) with JSON sidecars and Parquet metadata. Table 1 gives exact partition and class counts; hours are rounded to two decimals.

Split	Rows	Hours	BF	Spoof
Train	22,500	59.56	7,500	15,000
Development	21,000	52.69	6,952	14,048
T1 dev-test	16,023	40.94	4,097	11,926
T1 final	144,210	366.16	36,903	107,307
T2 dev-test	14,193	36.51	3,510	10,683
T2 final	127,746	329.03	31,490	96,256
Total	**345,672**	**884.88**	**90,452**	**255,220**

Table 1: Canonical published partitions. BF denotes bona fide; test labels were hidden during the challenge and released afterwards.

4.2 Source Collections and Eligibility

The manifests combine ADI-17/ADI-20 and MGB-lineage speech (Shon et al., 2020; Ali et al., 2019; Elleuch et al., 2025) with organizer-held collections. MSA draws from audiobooks, news anchors, Tunisian MSA, MSA YouTube, and MediaSpeech (Kolobov et al., 2021); Egyptian from an internal Egyptian collection and ADI-17; and Gulf from Saudi podcasts, ADI-17, and Bahraini YouTube. Test-only Levantine draws from Lebanese, Jordanian, and Syrian recordings, Jordanian YouTube, and ADI-17. Test-only Maghrebi draws from Moroccan and Tunisian recordings, ADI-17, and TunSwitch (Ben Abdallah et al., 2024). Every selected utterance retains its source in the organizer manifest.

Following the broad MGB-3 convention (Ali et al., 2017), manifest codes were grouped as MSA; Egyptian (EGY); Gulf (KSA, UAE, KUW, QAT, BAH, OMA, YEM, and IRA); Levantine (JOR, LEB, SYR, and PAL); or Maghrebi (TUN, MOR, MAU, ALG, and LIB). An utterance was eligible if it had UTMOS ≥ 3.0 (Saeki et al., 2022), a nonempty Zipformer-XL ASR transcript (Yao et al., 2024), and duration in $[0, 15)$ seconds; the transcript gate removed 88 records. Core-family sampling used seed 42 and preserved the available proportions of the $[0, 5)$, $[5, 10)$, and $[10, 15)$-second bands.

The test-only families used equal band quotas with backfill. Levantine contributed 834/833/833 utterances across the three bands. Maghrebi supplied all 387 eligible short utterances and 1,057/1,056 from the longer bands. Levantine comprises 1,294 JOR, 1,012 LEB, 141 SYR, and 53 PAL records; Maghrebi comprises 1,438 TUN, 629 MOR, 313 MAU, 89 ALG, and 31 LIB records. Levantine sources comprise 972 Lebanese, 631 Jordanian, 62 Syrian, 658 Jordanian YouTube, and 177 ADI-17 recordings. Maghrebi sources comprise 604 Moroccan, 420 Tunisian, 1,018 TunSwitch recordings, including 42 code-switched items, and 458 ADI-17 recordings.

4.3 Role-Based Source Pools

Figure 2 shows the two-lane construction. Pool 1 supplies bona fide speech. Pool 2 contains real recordings whose transcripts and voices serve as targets for synthesis. Levantine and Maghrebi are reserved for Track 1. The eligible inventories contain 11,220 MSA, 18,079 Egyptian, 72,037 Gulf, 8,875 Levantine, and 6,395 Maghrebi utterances. Each core pool has 77,500 records: MSA contributes 2,500 to each source split, Egyptian 2,500/7,500/7,500, and Gulf 2,500/25,000/25,000 for train/development/test. Each test-only family contributes 2,500 records.

Pool 2 uses EGY for Egyptian synthesis and KSA for Gulf synthesis. Pool 1 prioritizes non-Saudi Gulf recordings, but KSA still accounts for 70,461 of the 72,037 eligible Gulf records. A direct comparison of utterance IDs shows that Pool 1 and Pool 2 overlap by 1,259/1,260/1,261 MSA records, 2,431/7,294/7,296 Egyptian records, and 1,249/15,858/15,856 Gulf records across train/development/test. This planned overlap supplies matched real and synthetic examples; the remaining Pool 2 records add unpaired targets. Saudi podcasts supply 69,850 Gulf records, while MSA includes 6,693 audiobook and 2,292 news-anchor records.

4.4 Source-Linked Synthesis

Each synthetic utterance uses one Pool 2 record: its ASR transcript provides the text and its audio provides the cloning reference. References underwent Resemble Enhance (Resemble AI, 2023), loudness normalization, and model-rate resampling; supported systems also received Arabic family or dialect hints. Pool overlap creates matched real–synthetic cases, while the remaining targets add unpaired synthetic examples.

The internal HumainVoice-Internal-TTS system occurs in every split, as does VoxCPM2 (Zhou et al., 2026). F5-TTS, Fish Speech 1.5, and ElevenLabs (Chen et al., 2024; Liao et al., 2024; ElevenLabs, 2026) are test-only: F5-TTS covers Gulf, Fish Speech covers Egyptian and MSA, and ElevenLabs covers all five families. Withholding generated ElevenLabs train/development material preserves it as a held-out system. Table 2 reports final challenge rows. Train has 7,500 bona fide and

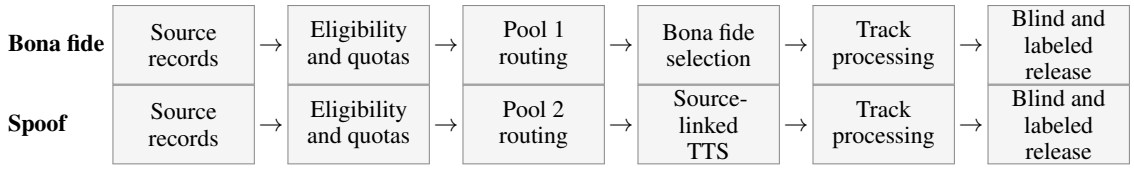


Figure 2: Two-lane construction process. Numerical split, generator, dialect, and condition composition is kept in Tables 1–2 rather than duplicated in the diagram.

7,500 utterances from each seen synthesizer; development has 6,952, 7,045, and 7,003, respectively. Both seen synthesizers re-synthesize 4,939 training sources also released as bona fide.

Track 1 additionally contains 2,000 records imported from MLAAD (Müller et al., 2024): 1,000 spoofed and 1,000 bona fide. The imported component pairs generated speech with English M-AILABS audiobook recordings, adding a focused cross-lingual condition. Its bona fide side contributes 87 development-test and 913 final-test trials. The 1,000 spoofed trials form one MLAAD source category for evaluation and reporting.

4.5 Test Assembly and Track Processing

The Track 1 pool contains 160,233 rows: 16,023 development-test and 144,210 final-test. Track 2 contains 141,939 rows: 14,193 and 127,746, respectively. We partitioned released trials rather than underlying source utterances to retain broad generator, family, and condition coverage in both phases. Thus, 12,866 Track 1 source IDs appear in both through different trials.

Every Track 1 item passes through one logical-access (LA) transform followed by one deepfake-channel (DF) transform. The six LA choices are 8-kHz A-law; a 300–3,400-Hz PSTN simulation with gain in $[-3, 3]$ dB followed by an A-law/GSM tandem; 16-kHz G.722; 8-kHz μ-law; GSM full-rate; and Opus at 8, 12, or 16 kbps (Valin et al., 2012). The eight DF choices are MP3 at 32/128 kbps, M4A at 32/128 kbps, Ogg Vorbis at quality 0/6, MP3-32→M4A-64, and Ogg-q0→M4A-64. All 48 LA–DF combinations occur, with near-balanced LA and DF marginals. Bona fide and spoofed speech receive the same processing, so the track tests detection after a complete transmission-and-compression chain.

Track 2 assigns exactly one condition to each item: clean (28,388 rows), codec round-trip (37,850), additive noise (37,851), or reverberation (37,850). Codec conditions use M4A at 20–32 or 96–112 kbps, MP3 at 80–120 or 220–260 kbps, and Ogg Vorbis at quality 0 or 6. Noise uses 930 MUSAN noise recordings, excluding music and babble (Snyder et al., 2015), at an SNR in $[0, 15)$ dB; 34,325 assignments use FreeSound recordings and 3,526 use SoundBible recordings. Reverberation uses 4,094 room impulse responses: 12,366 small-room, 12,425 medium-room, 12,759 large-room, and 300 real-RIR assignments from RIRS_NOISES/OpenSLR 28 (Ko et al., 2017). Together, they cover common coding, noise, and room-acoustic effects.

Dimension	Category	Train	Dev	T1 test	T2 test
Source family	MSA	7,500	1,459	10,647	10,647
	Egyptian	7,500	4,529	30,645	30,645
	Gulf	7,500	15,012	100,647	100,647
	Levantine	0	0	8,147	0
	Maghrebi	0	0	8,147	0
	MLAAD	0	0	2,000	0
Class/system	Bona fide	7,500	6,952	41,000	35,000
	HumainVoice-Internal-TTS	7,500	7,045	40,000	35,000
	VoxCPM2	7,500	7,003	40,000	35,000
	F5-TTS	0	0	25,000	25,000
	Fish Speech 1.5	0	0	9,998	9,998
	ElevenLabs	0	0	3,235	1,941
	MLAAD spoof	0	0	1,000	0
Test condition	T1 LA×DF	–	–	160,233	–
	T2 clean	–	–	–	28,388
	T2 codec	–	–	–	37,850
	T2 noise	–	–	–	37,851
	T2 RIR	–	–	–	37,850

Table 2: Final composition by separate descriptive axes. T1/T2 columns combine development-test and final-test; the panels describe the same rows. The MLAAD source family contains 1,000 bona fide and 1,000 spoofed trials.

4.6 Packaging, Blindness, and Integrity

The release uses 88 WebDataset shards of at most 4,096 FLAC/JSON pairs. Parquet and compressed-CSV indexes record paths, audio properties, SHA-256 hashes, and byte sizes. Hugging Face exposes nine configurations: `train_dev`, four challenge-era blind tests, and four labeled mirrors. Since `train_dev` contains train and development, the loader exposes ten splits.

Train/development names include family and a predicted-gender token. Test names follow `test_<7-digit-id>` and hide label and provenance. Each labeled mirror preserves its blind split's IDs, order, paths, hashes, and durations,

adding only the label. Test IDs are configuration-local: 203 recur across development-test configurations and 2,013 across final-test configurations, so the record key is `(configuration, id)`.

The organizer mapping connects all 345,672 records to their source, transcript, family, generator, and condition. Validation confirmed unique record keys, agreement between metadata and shard membership, uniform 16-kHz mono audio, and exact ID/path/duration/hash correspondence between blind and labeled configurations. The mapping supports reproducible scoring and the subgroup analyses reported here.

5 Evaluation

5.1 Metric

Both tracks are ranked by **Equal Error Rate (EER)**, the standard anti-spoofing metric: the point where the false acceptance rate (spoofed audio accepted as genuine) equals the false rejection rate (genuine audio rejected as spoofed) as the decision threshold varies.

5.2 Submission and Leaderboards

Both tracks were hosted on Codabench, which scored submissions automatically and maintained a live leaderboard. Participants uploaded a CSV of `audio_id,score` pairs covering every utterance in the relevant partition; malformed or incomplete files were rejected by the scoring program. Resources, dataset, baseline checkpoint, scoring script and submission templates, were consolidated on the Hugging Face organization page so that participants could reproduce the official metric locally before submitting.

Participation was substantial. Track 1 attracted 35 registered participants who made 616 scored submissions across the two phases, and Track 2 attracted 30 participants and 419 submissions. Since the evaluation phase capped submissions per team, the great majority of these were development-phase runs, roughly 18 and 14 per participant respectively.

6 Baseline

We provide a single baseline for both tracks, chosen to represent the prevailing recipe in the anti-spoofing literature rather than to be competitive: an **XLS-R 300M** front end (Babu et al., 2022) paired with an **AASIST** graph-attention back end (Jung et al., 2022), trained on the official training split with no augmentation. The front end is fully fine-tuned; the back end reads a single SSL layer and produces a two-class output, and the submitted score is the difference between the bona fide and spoof logits. Inputs are cropped or padded to 64,600 samples (≈ 4.04 s) at 16 kHz.

The baseline is implemented in **DeepFense** (El Kheir et al., 2026a), an open-source PyTorch toolkit in which the front end, back end, loss and augmentation are interchangeable components specified through a single configuration file. We chose it so that participants could vary one component at a time without re-implementing the surrounding pipeline, and so that our own setup is fully recoverable from the released configuration. The checkpoint, the YAML configuration and a step-by-step tutorial for reproducing the reported numbers and extending the recipe are available at `https://huggingface.co/ArabicSpeech/ArA-DF-Baseline`. Two participating teams built their submissions directly on DeepFense.

The baseline reaches 14.54% EER on the Track 1 final test and 27.11% on Track 2 (14.65% and 27.21% respectively on the corresponding development-test partitions). Two observations are worth recording. First, the gap between the two tracks is large: the same system loses roughly 13 points of EER when moving from dialect shift to acoustic shift, confirming that unaugmented training keys on channel characteristics that the Track 2 degradations destroy. Second, the baseline is beaten by a wide margin, by more than an order of magnitude at the top of both leaderboards, and, as Section 8 shows, most of that gain is attributable to the data pipeline rather than to the architecture.

7 Results

Table 4 first characterizes the eight systems with description papers along the axes that distinguished them (front end, adaptation strategy, back end, augmentation and fusion scheme), with the comparison across those axes taken up in Section 8.

Tables 5 and 6 give the official evaluation-phase leaderboards.[1] Fourteen teams were ranked on Track 1 and twelve on Track 2, with ten appearing on both.

[1] Some system description papers report slightly different ranks or team totals, having been written against an earlier leaderboard snapshot; the tables here reflect the final official standings.

Team	Leaderboard name	Affiliation	Track 1	Track 2
Raid AI	Raid AI	Raid AI, Egypt	✓	
SparkShield	DF-SAI	Anhui SparkShield / Anhui Univ. / iFLYTEK, China		✓
pi-labs.ai	pi-labs-ai	pi-labs.ai, India	✓	
UZH-CL	UZH-CL	Univ. of Zurich, Switzerland; Idiap; BUT, Czechia	✓	✓
R_A	DeepFake_Detector	Independent; HIAST, Syria	✓	✓
SaAr-DF	abdelrahmansakr	Alexandria University, Egypt	✓	
Thakaa	Thaka	Thakaa Advanced Company for IT, Saudi Arabia	✓	✓
PogranichnyZashchitnik	Minh-Hoang Le	Univ. of Information Technology, VNU-HCM, Vietnam	✓	✓

Table 3: Teams that submitted a system description paper

Team	Front end(s)	Adapt.	Back end	Augmentation	Fusion	EER T1 / T2
Raid AI	XLS-R 0.3/1/2B	full	TCM+ASP; AASIST	codec sim., RIR, noise	3-crop TTA	**0.87** / 1.32
SparkShield	XLS-R 0.3+1+2B	full	MultiConv+MHAP	RIR/noise, codec, gain	3-model score	0.88 / **0.66**
pi-labs.ai	XLS-R, WavLM-L	full	AASIST	telephony, pseudo-label	z-norm, 2-model	0.93 / 14.40
UZH-CL	W2V-BERT-2.0 (+3)	WPT $<1\%$	x-layer attn.+ASP	DSU, RawBoost, noise/RIR	track-adaptive, 15	1.96 / 1.04
R_A	WavLM-L, XLS-R, HuBERT	full	attentive pooling	50% degradation chain	LR meta-clf.	2.19 / 7.21
SaAr-DF	AntiDeepfake 0.3/1/2B	full	mean-pool *vs.* AASIST	RawBoost	13-member logit	2.42 / —
Thakaa	XLS-R $\times 4$, ResNet $\times 2$	full	stats pool.; AASIST	5–9 cond. classes, RIR	rank-norm, 2-stage	2.90 / 2.76
PogranichnyZ.	WavLM-L	LoRA 1.7M	ASP+OC-Softmax	codec, noise, RIR, SpecAug	none	12.54 / 5.66

Table 4: The eight systems with description papers, ordered by Track 1

Rank	Team	EER (%)
1	Raid AI*	**0.87**
2	DF-SAI*	0.88
3	hassan111	0.89
4	pi-labs-ai*	0.93
5	Whispeak	1.37
6	UZH-CL*	1.96
7	DeepFake_Detector*	2.19
8	abdelrahmansakr*	2.42
9	Thaka*	2.90
10	farisalasmary	4.66
11	KANd	4.84
12	VoiceCodeAI	5.19
13	NAMAA	8.71
14	Minh-Hoang Le*	12.54
–	*Official baseline*	*14.54*

Table 5: Track 1 (Dialect Generalization) official leaderboard, 144,210 utterances. *: team with a system description paper.

Rank	Team	EER (%)
1	DF-SAI*	**0.66**
2	hassan111	0.97
3	UZH-CL*	1.04
4	Raid AI*	1.32
5	Whispeak	2.60
6	Thaka*	2.76
7	Minh-Hoang Le*	5.66
8	DeepFake_Detector*	7.21
9	farisalasmary	14.07
10	pi-labs-ai*	14.40
11	rowanmashaal	23.38
–	*Official baseline*	*27.11*
12	qadam-labs	33.53

Table 6: Track 2 (Acoustic Robustness) official leaderboard, 127,746 utterances; the baseline is placed in score order.

Four observations frame the discussion that follows.

Nearly every submission beat the baseline. All fourteen ranked Track 1 teams finished below the 14.54% baseline, and eleven of twelve did so on Track 2, with a single entry at 33.53% falling short. The margins at the top are an order of magnitude: 0.87% on Track 1 and 0.66% on Track 2.

The top of each track is a statistical tie. On Track 1, ranks one to four span 0.055 percentage points (0.8719 to 0.9265 at the four-decimal precision reported by Ehab et al. (2026)), and the margin between first and second is 0.0088. On Track 2 the top four span 0.66 points, but every adjacent gap within that group is below 0.35. Both are small relative to the run-to-run variance Raid AI estimate from three seeds, a standard deviation of 0.25 points and a range of 0.47 (Ehab et al., 2026). We therefore describe the leading systems on each track as competitive with one another rather than strictly ordered, and regard the ranking as informative from roughly rank five downward, where the gaps exceed a point.

The two tracks do not rank systems the same way. Ten teams appear on both leaderboards, and their rankings correlate at Spearman $\rho = 0.58$: related, but far from interchangeable. The movements are large, pi-labs.ai placing fourth on Track 1 at 0.93% but tenth on Track 2 at 14.40% and Minh-Hoang Le moving the opposite way, from last to seventh; only two teams hold the same relative position on both. This is the clearest support for

separating the two conditions: a pooled evaluation would have averaged these movements away.

Acoustic shift is severe when untreated and tractable when addressed. Track 2 starts harder than Track 1 by baseline (27.11% vs 14.54%) but ends easier at the top (0.66% vs 0.87%), because augmentation closes the acoustic gap directly (Fang et al., 2026), giving Track 2 a wider spread than Track 1. Dialect shift has no such fix: PogranichnyZashchitnik's heavily-augmented system scores 5.66% on Track 2 but 12.54% on Track 1, with 11.5% of Track 1 spoofs scored as confidently bona fide (Le, 2026). UZH-CL find the converse, their dialect-specialized detector *raising* EER in the Track 2 fusion (Farhadipour et al., 2026), so adapting to one shift can hurt the other.

8 Lessons from the First Arabic Speech Deepfake Challenge

Despite their methodological diversity, the eight systems converge on a small set of design choices. Participant-reported ablations quantify the effects of several of these choices, as summarized in Table 7. Because the teams used distinct end-to-end pipelines, the reported effects are derived from *within-system* ablations rather than controlled cross-team comparisons. Section 9 discusses the implications of this constraint.

Models. Every system builds on a pre-trained SSL encoder, XLS-R in seven of eight, and half the field replaces the final layer with a learnable weighted sum over all layers (El Kheir et al., 2025c; Xiao and Vu, 2025). Three teams found AASIST beaten by a simpler pooling head, matching evidence that graph-attention back ends generalize poorly out of domain (Sakr and Elzahaby, 2026; Dowerah et al., 2026; El Kheir et al., 2026a). Parameter-efficient adaptation was competitive: UZH-CL's prompt tuning updates under 1% of parameters and places third on Track 2.

Data. Data was the largest lever in every ablation that isolated it, moving a fixed architecture from 13.13% to 1.37% in the clearest case (Ehab et al., 2026; Fang et al., 2026; Albani and Sonbol, 2026). The most useful negative finding also lives here: **adding external synthetic Arabic speech reliably hurt**, whether ArFake (Elsetohy et al., 2025), ArVoice (Toyin et al., 2025), copy-synthesis pseudo-spoofs (Alowisheq et al., 2024; Siuzdak, 2023) or MLAAD. The cause appears to be artifact rather than topical mismatch, and the harm scales with the mismatched fraction (Sakr and Elzahaby, 2026). A distinct use case was reported by UZH-CL, which used external corpora to increase diversity among ensemble components rather than to improve any individual model.

Design choice	Teams	Reported within-system effect (EER)
SSL encoder front end	8/8	universally adopted; no non-SSL comparison
Learnable layer aggregation	4/8	adopted; effect not isolated
Pooling head instead of AASIST	5/8	20.2→7.0; 5.2→4.4
Channel-matched augmentation	8/8	7.0→1.5; 6.3→2.3
Data curation (excluding OOD data)	–	2.6→1.5
External synthetic Arabic	4/8	degraded performance in 3 of 4 reported uses
Longer analysis window (4→10 s)	–	28.7→20.2
Multi-crop test-time augmentation	2/8	18.3% relative reduction in one study; track-dependent in another
Score fusion	7/8	≥40% relative reduction vs. the best component
Encoder scale (0.3B→2B)	3/8	≤0.09 percentage-point change

Table 7: Design choices reported across the eight system papers and their effects when evaluated through within-system ablations ((Fang et al., 2026; Sakr and Elzahaby, 2026; Albani and Sonbol, 2026; Ehab et al., 2026; Alhammad and Aldahlawi, 2026); Sections 7–8). Adoption counts denote the number of system papers reporting each choice; "–" indicates that the adoption count was not consistently reported. Reported effects are changes in EER within individual systems and should not be compared directly across rows.

Training and inference. Three cheap techniques paid off: longer analysis windows, multi-crop averaging at test time (which helps Track 1 but hurts Track 2 (Farhadipour et al., 2026)), and score fusion, used by seven of eight teams. Thakaa show why fusion works, contribution tracking error *independence* rather than individual accuracy (Alhammad and Aldahlawi, 2026).

Recommendations. The participant-reported ablations suggest five practical recommendations for future work on Arabic deepfake detection. (1) Prioritize channel-matched augmentation and training-data curation over increasing encoder scale. Data-focused interventions reduced EER by up to an order of magnitude, whereas encoder scaling changed EER by at most 0.09 percentage points in the reported comparisons. (2) Before incorporating an external synthetic corpus at scale, evaluate it at small mixture proportions and assess whether its generation artifacts are compatible with the target domain. Performance degraded in three of four reported uses, with larger degradation at higher proportions of mismatched data. (3) Select ensemble components based on error *complementarity*, as assessed using score correlation or error overlap, rather than standalone accuracy. (4) Use duration-aware handling for utterances shorter than the analysis window. Short utterances accounted for a dis-

proportionate share of the residual errors in every reported duration-based analysis. (5) Report variability across random seeds. The only reported multi-seed analysis had a standard deviation of 0.25 percentage points and a range of 0.47 percentage points in EER, exceeding all leaderboard gaps among the top four systems on both tracks. Differences of this magnitude from single runs should therefore not be interpreted as reliable evidence of method-level improvements.

9 Conclusion

We presented ArA-DF 2026, the first shared task on Arabic speech deepfake detection, with two tracks separating dialect generalization from acoustic robustness. Training data and augmentation, rather than architecture, drove the order-of-magnitude gains over the baseline, and the two tracks ranked systems differently ($\rho = 0.58$), supporting their separation. Short utterances and dialect shift remain open; a second edition should add duration-stratified evaluation with duration-adaptive scoring, broader dialect and generator coverage under leave-one-generator-out protocols (Elsetohy et al., 2025), source tracing, localization (Zhang et al., 2023), explanation (El Kheir et al., 2026b), and group-wise fairness and calibration reporting (El Kheir et al., 2026a).

Limitations

The eight systems differ simultaneously in their encoders, back-end architectures, training data, augmentation strategies, and fusion methods; cross-team score differences therefore cannot be attributed to individual techniques. Accordingly, claims in Section 8 about the effects of specific techniques rely on participant-reported within-system ablations. With one exception, these results are based on single runs. The sole multi-seed analysis reported a standard deviation of 0.25 percentage points and a range of 0.47 percentage points in EER, suggesting that differences below approximately 0.5 percentage points should be interpreted cautiously. Duration correlates with the label in every partition (clips under 2 s are 7.7% bona fide against 25.6% overall), so the per-duration error rates we cite assume skewed priors, and the labeled data is over 80% male. We rank by EER alone, which says nothing about calibration, and did not regulate transductive use of the evaluation audio, which two teams made use of (P and Iyengar, 2026; Farhadipour et al., 2026).

Ethics Statement

Speech deepfake detection is dual-use, but release is favorable: the generation systems are already public whereas robust Arabic detection resources were not. The bona fide speech comes from publicly available online media (Shon et al., 2020; Ali et al., 2019; Elleuch et al., 2025), and the spoofed partition clones a subset of it without individual consent, which we judge proportionate for an anonymized, research-only release built to defend against this misuse. A reported EER below 1% is not a deployment guarantee, and since error is dominated by spoofs accepted as bona fide, evidentiary use should keep a human in the loop.

Acknowledgments

We thank all participating teams for their submissions and system descriptions, and the ArabicNLP 2026 organizers for hosting the shared task. We thank Feidi Kallel for the reviewing support.

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